

Yulian Zhu-Yu

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Skills

- **Languages:** Python, C++, Java, Typescript
- **CI/CD:** Servers, Github, Docker
- **CAD:** Inventor, Fusion, Onshape
- **Data Visualization:** pandas, NumPy, Matplotlib
- **Security:** Web Exploitation, Reverse Engineering, GDB
- **OS & Tools:** Linux (Ubuntu), SSH, CLI
- **Robotics:** ROS2, Gazebo, Rviz2

Previous Experience

Computer Specialist

July 2025 - August 2025

- Remotely interfaced with servers through **PuTTY** and **Powershell**
- Developed a **Wordpress** website from scratch utilizing a website builder and **PHP**
- Personally created 3 plugins for add-on website functionality
- Migrated over 3400 products between **SQL** and **Excel** databases
- Learnt command line tools and **Linux**
- Made **custom macros** to edit over 5000 images

Robotics Club Mentor/Senior

January 2022 - July 2025

3388C - Cortisol

Calgary, Alberta

- **Built a championship robot** over multiple **iteration cycles**
- Sole programmer for the robot using **C++** for **real time control** and **external libraries**
- Worked in a team of 6 for design decision and management
- Handled **logistics** for **ordering parts** and splitting costs as well as inventory

Projects

Kobuki Autonomous Robot

October 2025

- Utilized **ROS2** along with **Nav2** to set up 3D mapping
- Implemented **Computer Vision** with a **Kinect camera** as a RGBD sensor
- Interfaced with a **Raspberry Pi**, between Kinect and the **Kobuki Base**
- Built robot perception and navigation in simulation using Gazebo and Rviz2

Homelab

- Dell optiplex running headless **Ubuntu Server 26.04**
- Multiple programs running locally using **Docker**
- Proxy using **Nginx** proxy manager
- VPN using **Tailscale**
- Docker and **Python flask** to set up local websites

Capture the Flag Competitions

- PicoCTF, PascalCTF, NOPSctf: Completed **Web Exploitation** and **Reverse Engineering**
- Placed in the top **100** in PascalCTF.

Education

Honours Bachelor of Science in Computer Science

GPA: 3.56

University of Toronto Mississauga

- Operating Systems & Tools: Linux (Ubuntu), SSH, command-line interface
- Python, Algorithms and Data Structures, OOP
- Unit Tests: Pytest, Hypothesis, Pyta